USC Viterbi
School of Engineering

Data Science Capstone Project
Deadline: 2025.May 10th

1. Project Goal

In this project, you will design your own interative data science dashboard.

You are expected to identify a meaningful problem, structure it into analytical questions, select appropriate quantitiative methods, and communicate your findings through an interactive dashboard.

The primary objective is not simply to build a sophisticated model or produce attractive visualizations. Instead, you should demonstrate your ability to transform a real-world problem(or virtual, but highly realistic) into an analytical framework, and derive meaningful, actionable insights from data.

Your project should adress four fundamental questions:

1. What? What problem are you trying to solve, and why does it matter?
2. Why? What factors, mechanisms, or relationships may explain the problem?
3. How? How can data science methods help investigate or solve the problem?
4. So What? What conclusions, recommendations, or decisions can be derived from your analysis?

You have substantial freedom in choosing your topic, dataset, analytical methods, and dashboard architecture.

Key takeaways: Problem Framing, Analytical Modeling, Interactive Design

2. Topic, Problem structure, and Skill

2.1. Topic selection

You are free to select any topic, industry, or problem that interests you.

Exmaples include, but are not limited to:

- Business strategy, marketing, and customer analytics
- Finance, investment, and risk management
- Logistics, operations, and product opimization
- Economics, public policy, and social issues
- Healthcare, education, and environmental analytics
- Technology, digital platforms, and user behavior.

Your topic should be sufficiently specific to support a clear problem definition and a feasible analytical investigation.

2.2. Problem structure

Define your project around one or more analytical objectives.

Possible objective include:

Problem Type	Example analyical question
Optimization	How can limited resources be allocated efficiently?
Prediction	What will happen under a given set of conditions?
Classification	Which category does an observation belong to?
Clustering	Can meaningful patterns or groups be identified?
Explanation	What factors are associated with the observed outcome?
Decision support	Which alaternative should be selected, and why?

You may combine multiple objectives if they contribute to the same overarching problem.

2.3. Analytical approaches

Select analytical approaches that are appropriate for your problem.

(Optional) You may use one or more of the following:

Type of Analytics	Description
Root Cause Analysis	Identify potential underlying causes of an observed problem
Causal Inference	Estimate the causal effect of an intervention, treatment, or policy.
Counterfactual Analysis	Examine how outcomes might differ under alternative assumptions or interventions.
Survival Analysis	Analze time-to-event outcomes, including customer churn, equipment failure, or default.

Decision Analysis	Evaluate alternatives under uncertainty, costs, benefits, and constraints.
Scenario and Sensitivity Analysis	Examine how analytical results change, when assumptions or external conditions vary.

You may also propose alternative approaches, provided that you explain their relevance to your problem.

2.4. Technical skills

Your project **must** demonstrate the application of **at least one** of the following techinical areas:

Type of skills	Description
Mathematical Programming	Linear programming, Mixed-integer programming, or other mathematical optimization techniques.
Machine Learning	Supervised learning, Unsupervised learning, or statistical learning methods.
Deep Learning	Neural networks, transformers, or other deep learning architecture.

You may combine multiple techniques approaches.

However, model complexity alone will not determine your grade. Your choice of methods **must** be justified by the analytical objective, available data, assumptions, and computational requirements.

3. Requirements

3.1. Problem Definition

Your project **must** contain:

1. Problem statement: Clearly articulate the problem you intend to investigate or solve.
2. Motivation and Significance: Explain why the problem matters and who would benefit from solving it.

3. Problem Decomposition: Break the overarching problem into at least three specific analytical questions or subproblems.
4. Analytical Objectives: Define the expected analytical outputs, such as predictions, explanations, segmentations, or optimized decisions.
5. Success Criteria: Establish measureable criteria for evaluating whether your analysis successfully addresses the problem.

You may use a Problem Tree, Issue Tree, MECE framework, or another structured problem-framing technique.

3.2. Data requirements

Your project **must** use a dataset suitable for the selected analytical objectives.

You are required to:

- Describe the data source, collection period, observation units, and key variables.
- Explain the data acquisition and preprocessing precedures.
- Identify missing values, outliers, data leakage risks, and potential sampling biases.
- Justify the relevance of the selected features to your analytical questions.
- Discuss limitations of the dataset and their implications for interpretation.

You may use public datasets, APIs, web-collected data collected in accordance with applicable terms, or appropriately documented synthetic datasets.

If synthetic data is used, clearly distinguish simulated patterns from real-world evidence.

3.3. Analytical and Modeling Requirements

Your project **must** implement **at least one** mathematical programming, machine learning, or deep learning method.

You are required to:

1. Explain the theoretical or mathematical logic behind your selected method.
2. Specify the key assumptions, variables, parameters, and expected outputs.
3. Establish an appropriate baseline or benchmark.
4. Evaluate the method using relevant performance metrics or analytical validation procedures.
5. Interpret the results in relation to the orignal problem definition.

Where appropriate, perform sensitivity analysis, robustness checks, or comparisons across alternative models.

Important: Predictive associations must not be presented as causal effects without an appropriate identification strategy.

3.4. Dashboard requirements

Your dashboard **must** contain **at least four distinct analytical layers or sections**, serving different analytical purposes.

A recommended architecture is:

Layer	Purpose	Possible components
1.Overview	Describe the overall situation and key performance indicators.	KPI cards, descriptive charts, summary metrics
2.Exploration	Investigate patterns, distributions, and relationships	Distribution plots, correlation charts, segmentation views
3.Modeling	Present model outputs, validation results, and performance	Prediction charts, model comparison, feature importance
4.Decision Support	Translate analytical results into decisions or scenarios	Scenario simulators, optimization results, sensitivity charts

You may reorganize these layers based on your project's analytical needs.

Minimum visualization requirements:

- At least four meaningful chart-based visualization across the dashboard.
- At least two interative chart whose contents update in response to user-selected inputs.
- At least one interactive analytical control, such as a parameter slider, scenario selector, model selector, or filtering mechanism.
- Clear chart titles, axes, legends, units, and contextual explanations.
- A logical navigation structure that helps users follow the analysis.

An interactive chart **must** change its displayed analytical information based on user input. Simple zoming, hovering, or displaying a static tooltip does not satisfy this requirement.

You may use Netlify, Streamlit, Dash, Plotly, Shiny, or other suitable dashboard development frameworks.

3.5. Reproducibility and Documentation

Your project **must** be reproducible and sufficiently documented.

Include:

- A clear project directory structure and source code.
- Instructions for environment setup and dashboard execution.
- Data sources and acquisition procedures.
- Documentation of important functions, algorithms, and model configurations.
- Model evaluation results and known limitations.
- A README file describing the project's objectives, implementation, and usage.

You **must** ensure that submitted code can be executed using the documented environment and available data or reproducible data-acquisition procedures.

4. Tasks and Deliverables

The project consists of five stages.

Task 1. Define and Structure the problem

Develop a concise problem statement and explain its significance.

Your work should address:

- What is the central problem?
- Who are the relevant stakeholders?
- What are the suspected causes, constraints, or contributing factors?
- How can the problem be decomposed into analytical tasks?
- What specific outcomes would indicate a successful project?

Deliverable: Problem definition document, including a problem structure diagram and measurable success criteria.

Task 2. Data collection and Exploratory Analysis

Identify, acquire, and prepare the dataset.

Conduct exploratory data analysis to examine the characteristics of the data and identify potentially meaningful patterns.

Your analysis should include relevant descriptive statistics, data-quality checks, and exploratory visualizations.

Deliveralbe: Dataset documentation, preprocessing code, and exploratory analysis results.

Task 3. Develop the Analytical Model

Select and implement your analytical methods.

Explain why the chosen methods are suitable for the problem and discuss their underlying mathematical or statistical assumptions.

Compare your proposed method against a reasonable benchmark where applicable.

Assess the reliability and limitations of the results using appropriate validation procedures.

Deliverables: Working analytical model, methodological explanation, and evaluation results.

Task 4. Develop the Interative Dashboard

Design a dashboard that integrates exploratory findings, model outputs, and decision-relevant information.

Ensure that the dashboard has at least four analytical layers, four meaningful visualizations, and two interative charts.

Users should be able to navigate the analytical results and explort the implications of different inputs or assumptions.

Deliverable: Functional interative dashboard and execution instructions.

Task 5. Interpret Results and Recommend Actions

Prepare a final analytical report that connects your findings to the original problem.

Your discussion should address:

1. What are the most important findings?
2. How strongly does the evidence support your conclusions?
3. What decisions or actions could be informed by these findings?
4. How might the conclusions change under different assumptions?
5. What are the limitations, risks, and opportunities for future development?

Deliverables: Final analytical report and a dashboard demonstration

5. Grading

The total project score is **100 points**, allocated across three categories.

Visualization (25 points) :

Points	Criterion
8	Dashboard architecture: at least four distinct, logically organized analytical layers.
8	Visualization design: at least four clear and meaningful charts
5	Interactivity: at least two charts that update dynamically based on user input
4	Usability, navigation, labeling, and visual consistency
Subtotal: 25	

High-scoring dashboards should communicate analytical findings clearly, make efficient use of interactive controls, and help users understand relationships between variables.

Imporant: Visual complexity without analytical value will not receive additional credit.

Code and Mathematical logic (45 points) :

Points	Criterion
15	Technical implementation: correctness and appropriateness of the selected analytical methods.
12	Mathemetical and statistical reasoning: clear explanation of assumptios, algorithms, and model logic.
10	Evaluation and validation: appropriate benchmarks, performance metrics, and robustness assessments.

8	Code quality and reproducibility: Organization, Documentation, Reliable execution.
Subtotal: 45	

High-scoring proejcts should demonstrate strong technical reasoning and a clear understanding of **how and why** the selected analytical method work.

Students must be able to explain the mathemtical formulation, model behavior, and relevant limitations of their implementations.

Important: Using complex algorithms without understanding their underlying assumptions will not receive additional credit.

Quality of Analysis (30 points) :

Points	Criterion
10	Problem definition and structure: Clarity, Relevance, and Logical decomposition.
8	Analytical coherence: Alignment between the problem, data, methods, and findings.
8	Interpretation and actionable insights: Meaningful conclusions and Decision implications.
4	Critical thinking: Acknowledgement of uncertainty, limitations, and alternative explanations.
Subtotal: 30	

High-scoring projects should demonstrate that the student understands not only how to analyze data, but aslo why the analytis matters.

The project should establish a coherent logical connection:

Problem → Analytical Questions → Data → Methods → Evidence → Conclusions → Decisions

Imporant: A technically sophisticated model or visually attractive dashboard cannot compensate for an unclear problem deifinition, unsupported conclusions, or weak analytical reasoning.

6. General Evaluation Principles

The following principles apply across all grading categories.

Relevance over complexity: A relatively simple model that solves a well-defined problem can receive a higher grade than a complex model with limted practical relevance.

Evidence oveer speculation: Claims should be supported by appropriate data, transparent assumptions, or clearly identifed modeling results.

Interpretability over presentation alone: The dashboard must help users understand analytical findings rather than merely display statistical outputs.

Decision usefulness:Strong projects should explain how their findings can improve understanding, support decisions, or guide future action.

Intellectual independence: You are encouraged to explort unconventional questions, combine analytical approaches, and demonstrate creativity through rigorous problem formulation.

7. Submission

Checklist:

- Problem definition document
- Dataset and data document
- Reproducible source code
- Analytical model and evaluation
- Interactive dashboard
- README and execution instructions
- Final analytical report

Compress your all files into **.zip** and upload it through the Brightspace.